NOTRE DAME UNIVERSITY

BANGLADESH

CSE Assignment-01

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Assignment Topic: Quiz Questions Solve

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CSE-2103 Assignment No. 1

Quiz Questions:

1. **Consider the following infix expression:**
   a) 25+(5-1)-3/(2*7)
   b) 12/(7-3)+2*(1+5)
   Use Stack to translate the given expression into it's equivalent postfix Expression P.

2. **What is Stack...? Give a real-life example.**
   Do the following operation using Stack where MAX items in the Stack can be 6.
   PUSH: A, B, C, D, E
   POP;
   PUSH: F, G, H, I
   POP;

3. **Sort the Following Array using Bubble sort:**
   Array [ ] = {17, 51, 9, 39, 1, 6} **(Set B)**

4. **Find the element 19 in the following Array using Binary Search**
   Array [ ] = {5, 8, 9, 12, 15, 19, 25, 30, 35, 46, 51, 65, 80} **(Set A)**

CSE-2103 Assignment No. 1

Solution Part:

Answer to the Question No. 01 (a):

(25+(5-1)-3/(2*7))

Symbol Scanned	Stack	Expression P
a. (	(	
b. 25	(	25
c. +	(+	25
d. (	(+(	25
e. 5	(+(	25, 5
f. -	(+(-	25, 5
g. 1	(+(-	25, 5, 1
h.)	(+	25, 5, 1-
i. -	(-	25, 5, 1-+
j. 3	(-	25, 5, 1-+3
k. /	(-/	25, 5, 1-+3
l. (	(-/(	25, 5, 1-+3
m. 2	(-/(	25, 5, 1-+3, 2
n. *	(-/(*	25, 5, 1-+3, 2
o. 7	(-/(*	25, 5, 1-+3, 2, 7
p.)	(-/	25, 5, 1-+3, 2, 7*
q.)		25, 5, 1-+3, 2, 7*/-

Answer to the Question No. 01 (b):

(12/(7-3)+2*(1+5))

Symbol Scanned	Stack	Expression P
a. (	(	
b. 12	(	12
c. /	(/	12
d. (	(/(	12
e. 7	(/(	12, 7
f. -	(/(-	12, 7
g. 3	(/(-	12, 7, 3
h.)	(/	12, 7, 3-
i. +	(+	12, 7, 3-/
j. 2	(+	12, 7, 3-/2
k. *	(+*	12, 7, 3-/2
l. (	(+*(	12, 7, 3-/2
m. 1	(+*(	12, 7, 3-/2, 1
n. +	(+*(+	12, 7, 3-/2, 1
o. 5	(+*(+	12, 7, 3-/2, 1, 5
p.)	(+*	12, 7, 3-/2, 1, 5+
q.)		12, 7, 3-/2, 1, 5+*+

Answer to the Question No. 02:

Stack: A Stack is a list of elements in which an element may be inserted or deleted only at one end, called the top of stack.

Example: Phone Call List, Facebook Notifications, a branch of plates etc.

PUSH: To insert an element into a stack.

POP: To delete an element from a stack.

Here a Stack—MAX item = 6

a) Top = 0; PUSH(Stack, A)
When Top < MAX
Top = Top + 1
= 0 + 1
= 1

A

b) Top = 1; PUSH(Stack, B)
When Top < MAX
Top = Top + 1
= 1 + 1
= 2

B
A

c) Top = 2; PUSH(Stack, C)
When Top < MAX
Top = Top + 1
= 2 + 1
= 3

C
B
A

d) Top = 3; PUSH(Stack, D)
When Top < MAX
Top = Top + 1
= 3 + 1
= 4

D
C
B
A

e) **Top = 4; PUSH(Stack, E)**
When Top < MAX
Top = Top + 1
= 3 + 1
= 5

E
D
C
B
A

f) **Top = 5; POP(Stack)**
When Top ≠ 0
Top = Top - 1
= 5 - 1
= 4

D
C
B
A

g) **Top = 4; PUSH(Stack, F)**
When Top < MAX
Top = Top + 1
= 4 + 1
= 5

F
D
C
B
A

h) **Top = 5; PUSH(Stack, G)**
When Top < MAX
Top = Top + 1
= 5 + 1
= 6

G
F
D
C
B
A

i) Top = 6; PUSH(Stack, H)
When Top == MAX
STACK IS OVERFLOW
No Item cannot be inserted.
Return.

G
F
D
C
B
A

j) Top = 6; PUSH(Stack, I)
When Top == MAX
STACK IS OVERFLOW
No Item cannot be inserted.
Return.

G
F
D
C
B
A

k) Top = 6; POP(Stack)
When Top ≠ 0
Top = Top - 1
= 6 - 1
= 5

F
D
C
B
A

Answer to the Question No. 03:

Sorting the Following Array using Bubble sort:

Array [ ] = {17, 51, 9, 39, 1, 6}

Here Array Size n = 6;

1. When k = 1; k < n:
   a) **17, 51,** 9, 39, 1, 6
   b) 17, **51, 9,** 39, 1, 6
   c) 17, 9, **51, 39,** 1, 6
   d) 17, 9, 39, **51, 1,** 6
   e) 17, 9, 39, 1, **51, 6**
   f) 17, 9, 39, 1, 6, **51**

2. When k = 2; k < n:
   a) **17, 9,** 39, 1, 6, **51**
   b) 9, **17, 39,** 1, 6, **51**
   c) 9, 17, **39, 1,** 6, **51**
   d) 9, 17, 1, **39, 6, 51**
   e) 9, 17, 1, 6, **39, 51**

3. When k = 3; k < n:
   a) **9, 17,** 1, 6, **39, 51**
   b) 9, **17, 1,** 6, **39, 51**
   c) 9, 1, **17, 6, 39, 51**
   d) 9, 1, 6, **17, 39, 51**

4. When k = 4; k < n:
   a) **9, 1,** 6, **17, 39, 51**
   b) 1, **9, 6, 17, 39, 51**
   c) 1, 6, **9, 17, 39, 51**

5. When K = 5; k < n:
   a) **1, 6, 9, 17, 39, 51**
   b) **1, 6, 9, 17, 39, 51**

∴ So, the sorted Array [ ] = {1, 6, 9, 17, 39, 51}

Answer to the Question No. 04:

Finding the element 19 in the following sorted Array using Binary Search:
Array [ ] = {5, 8, 9, 12, 15, 19, 25, 30, 35, 46, 51, 65, 80}

Here Array size n = 13; Upper Bound is UB, Lower Bound is LB

int item = 19;
int MID = (LB+UB)/2;
int LOC = ?

1. When LB = 1; UB = 13; LB<=UB (true):
   a) MID = (1+13)/2 = 7;
   b) If (item == Array[MID]) →19 == Array[7] → (false)
   c) If (item < Array[MID]) → 19 < 25 → (true)
      UB = MID - 1 = 7-1 = 6;

2. When LB = 1; UB = 6; LB <= UB (true):
   a) MID = (1+6)/2 = 3;
   b) If (item == Array[MID]) → 19 == Array[3] → (false)
   c) If (item < Array[MID]) → 19 < 9 → (false)
   d) LB = MID + 1 = 3+1 = 4;

3. When LB = 4; UB = 6; LB <= UB (true):
   a) MID = (4+6)/2 = 5;
   b) If (item == Array[MID]) → 19 == Array[5] → (false)
   c) If (item < Array[MID]) → 19 < 15 → (false)
   d) LB = MID + 1 = 5+1 = 6;

4. When LB = 6; UB = 6; LB <= UB (true):
   a) MID = (6+6)/2 = 6;
   b) If (item == Array[MID]) → 19 == Array[6] → (true)
      LOC = MID;
      LOC = 6

∴ So, the element 19 Locate in index 6.